

CODEALPHA BIOINFORMATICS INTERNSHIP

TASK-1

Title-BLAST Analysis of Tumor protein p53 sequence

1. Introduction

Sequence analysis is an important bioinformatics approach used to identify relationships between biological sequences. BLAST (Basic Local Alignment Search Tool) compares a DNA or protein sequence against databases to identify homologous sequences and determine evolutionary relationships.

2. Protein Information

Protein name: Tumor antigen p53 (Tumor protein p53)

Organism: *Homo sapiens* (Human)

Database: UniProt

Accession ID: AAH03596

Length: 393 amino acids

Sequence type: Protein

FASTA Sequence:

```
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLS  
PDDIEQWFTEDPGPDEAPRMPEAA  
  
PRVAPAAPAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFR LGFLHSG  
TAKSVTCTYSPALNKMFCQLAKT
```

CPVQLWVDSTPPPGRVVRAMAIYKQSQHMTEVVRRCPPHHERCSDS
DGLAPPQHLIRVEGNLRVEYLDDRN
TFRHSVVVPYEPPEVGSDCTTIHYNMCMNSSCMGGMNRRPILTIITL
EDSSGNLLGRNSFEVRVCACAGR
DRRTEENLRKKGEPHHELPPGSTKRALPNNTSSSPQPKKKPLDGEY
FTLQIRGRERFEMFRELNEALEL
KDAQAGKEPGGSRAHSSHLKSKKGQSTSRRHKKLMFKTEGPDSD

3. BLAST Methodology

The sequence was analysed using NCBI BLAST. Protein BLAST (blastp) was performed against the non-redundant protein database. The resulting homologous sequences were evaluated based on percentage identity, query coverage, E-value, and alignment score.

An official website of the United States government [Here's how you know](#)

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Standard Protein BLAST

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query. more...

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Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) [Query subrange](#)

MEEPQSDPSVEPPLSQETFSDLWLLPENNVLSPLPSQAMDDLMLSPDDIE
QWFTEDPGPDEAPRMPEAA
PRVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGRLGFLHSGTAKSV
TCTYSPALNKMFCQLAKT

From
To

Or, upload file No file chosen [?](#)

Job Title
Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ?

Organism
Optional ☐ exclude
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Exclude ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences
Optional

Program Selection

Algorithm ☐ Quick BLASTP (Accelerated protein-protein BLAST)
☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)

[Feedback](#)

4.RESULT:

The BLAST results showed that the query sequence had the highest similarity with homologous proteins from closely related species. The top hit showed 98% identity and complete query coverage, indicating strong evolutionary conservation.

Descriptions		Graphic Summary	Alignments	Taxonomy				
Sequences producing significant alignments				Download	Select columns	Show	100	?
☐ select all 100 sequences selected		GenPept Graphics Distance tree of results Multiple alignment MSA Viewer						
	Description	Scientific Name	Common Name	Max Score	Query Cover	E value	Per. Ident	Acc. Len
☒	Tumor protein p53 [Homo sapiens]	Homo sapiens	human	816	100%	0.0	100.00%	393
☒	TP53_partial [synthetic construct]	synthetic construct	NA	815	100%	0.0	99.75%	393
☒	tumor protein p53 (Li-Fraumeni syndrome)_partial [synthetic construct]	synthetic construct	NA	813	100%	0.0	99.75%	393
☒	tumor protein p53 (Li-Fraumeni syndrome)_isoform CRA_c [Homo sapiens]	Homo sapiens	human	813	100%	0.0	99.75%	408
☒	TP53_partial [synthetic construct]	synthetic construct	NA	813	100%	0.0	99.75%	393
☒	p53 [Homo sapiens]	Homo sapiens	human	813	100%	0.0	99.75%	393
☒	tumor protein p53_partial [synthetic construct]	synthetic construct	NA	813	100%	0.0	99.75%	394
☒	TP53_partial [synthetic construct]	synthetic construct	NA	812	100%	0.0	99.75%	393
☒	TP53_partial [synthetic construct]	synthetic construct	NA	812	100%	0.0	99.75%	393
☒	tumor protein p53 (Li-Fraumeni syndrome) [Homo sapiens]	Homo sapiens	human	811	100%	0.0	99.49%	393
☒	p53 transformation suppressor_partial [Homo sapiens]	Homo sapiens	human	811	100%	0.0	99.49%	393
☒	p53 antigen [Homo sapiens]	Homo sapiens	human	811	100%	0.0	99.49%	393
☒	mutant p53 protein_partial [Homo sapiens]	Homo sapiens	human	811	100%	0.0	99.49%	393
☒	p53 transformation suppressor_partial [Homo sapiens]	Homo sapiens	human	811	100%	0.0	99.49%	393
☒	phosphoprotein p53 [Homo sapiens]	Homo sapiens	human	811	100%	0.0	99.49%	393
☒	tumor suppressor protein p53 [Homo sapiens]	Homo sapiens	human	811	100%	0.0	99.49%	393
☒	tumor protein p53 [synthetic construct]	synthetic construct	NA	810	100%	0.0	99.49%	393
☒	p53 transformation suppressor_partial [Homo sapiens]	Homo sapiens	human	810	100%	0.0	99.49%	393
☒	p53 transformation suppressor_partial [Homo sapiens]	Homo sapiens	human	810	100%	0.0	99.49%	393

5. Conclusion

BLAST analysis successfully identified homologous sequences for the selected protein. High sequence identity suggests that the protein is conserved among related organisms and may perform similar biological functions.

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